Primer: Maximum Causal Entropy IRL (MCE-IRL)

econirl package documentation

1 Problem and Motivation

Standard maximum entropy IRL (Ziebart et al., 2008) recovers a reward function from demonstrations by matching feature expectations under the maximum entropy principle. However, it uses non-causal state visitation frequencies that propagate future information backward, violating the agent's information structure. When features vary by action, this produces incorrect policies.

Ziebart (2010) introduced Maximum Causal Entropy IRL (MCE-IRL), which conditions each action only on information available at decision time. The causal entropy $H(A_t|S_{1:t}, A_{1:t-1})$ replaces the Shannon entropy $H(A^T)$, ensuring that the recovered policy respects the sequential timing of decisions. This is exactly what economists call "rational expectations" in dynamic programming.

The resulting softmax Bellman equation is simultaneously the MCE optimal policy (Theorem 1 of Ziebart 2010) and the structural logit choice model of Rust (1987). An economist sees structural estimation with feature matching. A machine learning researcher sees maximum entropy IRL. They solve the same optimization problem. This makes MCE-IRL the bridge estimator between the two traditions.

2 Model and Notation

The agent solves the maximum causal entropy problem

$$\max_{\pi} \; \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t \big(r(s_t, a_t; \theta) + \sigma \mathcal{H}(\pi(\cdot|s_t))\big)\right] \tag{1}$$

subject to the feature matching constraint $\mathbb{E}_{\pi}[\phi] = \mathbb{E}_{\text{data}}[\phi]$, where $r(s, a; \theta) = \theta^{\top}\phi(s, a)$ is linear in known features ϕ with unknown weights θ.

Symbol	Name	Definition
$Q(s, a)$	Soft Q-value	$r(s, a; \theta) + \beta \sum_{s'} P(s'\|s, a)\, V(s')$
$V(s)$	Soft value	$\sigma \log \sum_a \exp(Q(s, a)/\sigma)$
$\pi(a\|s)$	Policy (CCP)	$\exp\big((Q(s, a) - V(s))/\sigma\big)$
$D_{\pi}(s)$	State visitation	Expected discounted visits to s under π
ρ_0	Initial distribution	Empirical start-state frequencies
F_{π}	Policy transitions	$F_{\pi}[s, s'] = \sum_a \pi(a\|s)\, P(s'\|s, a)$
$\phi(s, a)$	Features	Known, shape $(\|\mathcal{S}\|, \|\mathcal{A}\|, K)$
θ	Reward weights	Unknown, dimension K

3 Derivations

3.1 Theorem 1: The MCE Optimal Policy

We seek the policy that is maximally noncommittal (highest entropy) while reproducing the observed feature statistics. The *causal* entropy conditions each action only on information available at decision time, respecting the agent's sequential information structure.

Primal problem.

$$\max_{\pi} \; H_{\text{causal}}(\pi) \quad \text{s.t.} \quad \mathbb{E}_{\pi}[\phi] = \bar{\phi}_{\text{data}}, \quad \pi(a|s) \geq 0, \quad \sum_{a} \pi(a|s) = 1 \; \forall s. \tag{2}$$

The causal entropy is

$$H_{\text{causal}}(\pi) = \sum_{t=0}^{\infty} \gamma^t \sum_{s} d_{\pi}^{t}(s) \sum_{a} \pi(a|s) \log \frac{1}{\pi(a|s)}, \tag{3}$$

where $d_{\pi}^{t}(s) = P(S_t = s \mid \pi)$ is the probability of being in state s at time t under policy π.

Lagrangian. Introducing multipliers $\theta \in \mathbb{R}^K$ for the feature constraints and $\lambda(s)$ for the normalization constraints,

$$(\pi, \theta, \lambda) = H_{\text{causal}}(\pi) + \theta^{\top}\big(\mathbb{E}_{\pi}[\phi] - \bar{\phi}_{\text{data}}\big) + \sum_{s} \lambda(s) \left(\sum_{a} \pi(a|s) - 1 \right). \tag{4}$$

Substituting the definitions and taking the derivative with respect to $\pi(a|s)$,

$$\frac{\partial}{\partial \pi(a|s)} = d_{\pi}(s) \Big[-\log \pi(a|s) - 1 + \theta^{\top} \phi(s,a) + \text{continuation} \Big] + \lambda(s) = 0. \tag{5}$$

The "continuation" term accounts for how changing $\pi(a|s)$ affects future state visitation. Solving and absorbing constants into $\lambda(s)$, we obtain

$$\pi^*(a|s) = \frac{\exp\big(Q^*(s,a)/\sigma\big)}{\sum_{a'} \exp\big(Q^*(s,a')/\sigma\big)}, \tag{6}$$

where Q^* satisfies the soft Bellman recursion

$$Q(s,a) = \theta^{\top} \phi(s,a) + \beta \sum_{s'} P(s'|s,a)\, V(s'), \tag{7}$$

$$V(s) = \sigma \log \sum_{a} \exp\big(Q(s,a)/\sigma\big). \tag{8}$$

The policy is therefore

$$\pi(a|s) = \exp\big((Q(s,a) - V(s))/\sigma\big). \tag{9}$$

This is exactly the structural logit model of Rust (1987) with Type I extreme value errors at scale σ. The Lagrange multipliers θ play the role of the structural parameters, and the soft Bellman recursion is the standard DDC value function.

3.2 Theorem 2: Feature Matching Gradient

Dual problem. Define the dual function $G(\theta) = \max_\pi(\pi, \theta)$. Since the inner maximization has the closed-form solution $\pi^*(\theta)$ from Theorem 1, we can write

$$G(\theta) = H_{\text{causal}}(\pi^*(\theta)) + \theta^\top \left(\mathbb{E}_{\pi^*}[\phi] - \bar{\phi}_{\text{data}}\right). \tag{10}$$

The MCE-IRL problem is $\min_\theta G(\theta)$.

Gradient by the envelope theorem. Since π^* is the unconstrained maximizer of for fixed θ, the envelope theorem gives

$$\nabla_\theta G(\theta) = \mathbb{E}_{\pi^*(\theta)}[\phi] - \bar{\phi}_{\text{data}} = \bar{\phi}_\pi(\theta) - \bar{\phi}_{\text{data}}. \tag{11}$$

The expected features under the current policy are computed via the occupancy measure

$$\bar{\phi}_\pi(\theta) = \sum_s D_\pi(s) \sum_a \pi(a|s)\, \phi(s, a), \tag{12}$$

where $D_\pi(s)$ is the expected discounted state visitation frequency under π. This satisfies the fixed-point equation

$$D_\pi = \rho_0 + \beta\, F_\pi^\top\, D_\pi \quad \Longrightarrow \quad D_\pi = (I - \beta\, F_\pi^\top)^{-1}\, \rho_0, \tag{13}$$

with $F_\pi[s, s'] = \sum_a \pi(a|s)\, P(s'|s, a)$ and ρ_0 the initial state distribution.

At the optimum, $\nabla_\theta G = 0$, so $\bar{\phi}_\pi = \bar{\phi}_{\text{data}}$. The gradient vanishes when the policy's feature expectations match the data.

Equivalence to NFXP. The conditional log-likelihood of the DDC logit model is $\ell(\theta) = \sum_i \log \pi_\theta(a_i|s_i)$. Its gradient is

$$\nabla_\theta \ell(\theta) = \frac{1}{\sigma}\left(\bar{\phi}_{\text{data}} - \bar{\phi}_\pi(\theta)\right), \tag{14}$$

which differs from Equation (11) only by the factor $1/\sigma$. MCE-IRL and NFXP optimize the same objective and converge to the same $\hat{\theta}$.

3.3 Theorem 3: Robustness Guarantee

The MCE distribution P_{MCE} solves a minimax problem over the set of distributions matching the feature constraints.

Statement. Among all distributions in , the MCE distribution minimizes the worst-case KL divergence to the true (unknown) distribution P':

$$P_{\text{MCE}} =_{P\in} \max_{P'\in} D_{\text{KL}}(P' \| P). \tag{15}$$

Proof sketch. The KL divergence $D_{\text{KL}}(P' \| P) = \mathbb{E}_{P'}[\log P'/P]$ is minimized (in P) when P places as much probability mass as possible on all outcomes consistent with the constraints. The maximum entropy distribution does exactly this: it is the "flattest" distribution in , making it the centroid of the constraint set in the KL geometry. Any other distribution $P \in$ concentrates mass away from some region, creating a worst-case P' that concentrates mass precisely in that region, yielding higher KL divergence. For a formal proof, see Chapter 3 of Ziebart (2010).

Practical consequence. This means the MCE policy hedges against adversarial changes in the state distribution. NFXP, which solves the same optimization, inherits this guarantee. However, the guarantee applies only within the constraint set defined by the chosen features ϕ. Misspecified features can still produce poor out-of-sample predictions.

3.4 Identification

Rewards are identified only up to potential-based shaping (Ng et al., 1999). For any $h : \mathcal{S} \to \mathbb{R}$, the shaped reward $r'(s, a) = r(s, a) + h(s) - \beta \sum_{s'} P(s'|s, a)h(s')$ produces the same optimal policy (Magnac and Thesmar, 2002). The parametric restriction $r = \theta^\top \phi$ with $K < |\mathcal{S}|(|\mathcal{A}| - 1)$ resolves this by constraining the reward to a low-dimensional subspace. MCE-IRL additionally constrains $\|\theta\|$ via the feature matching dual, selecting the maximum-entropy-compatible reward from the equivalence class (Kim et al., 2021).

4 Estimation Algorithm

Algorithm 1: MCE-IRL (Ziebart, 2010)

Input: Panel $\{(s_i, a_i)\}_{i=1}^N$, features $\phi(s, a)$, transitions $P(s'|s, a)$, discount β, scale σ

`// Step 1: Compute empirical feature expectations`
$\bar{\phi}_{\text{data}} = \frac{1}{N} \sum_{i=1}^N \phi(s_i, a_i)$
Compute empirical state occupancy $D_{\text{data}}(s) = \frac{1}{N} \sum_i \mathbf{1}[s_i = s]$
Compute initial state distribution ρ_0 from first observations

`// Step 2: Initialize reward weights`
$\theta_0 \leftarrow \mathbf{0}$

`// Step 3: Iterate until convergence`
for $k = 0, 1, \ldots$ **do**

`// 3a. Backward pass: soft value iteration (Eqs. 7--9)`
$R \leftarrow \theta_k^\top \phi$ `// reward matrix` $(|\mathcal{S}|, |\mathcal{A}|)$
Solve $V = T_\sigma(V; R)$ via hybrid iteration to get V, Q, π

`// 3b. Forward pass: occupancy measure (Eq. 13)`
$F_\pi[s, s'] = \sum_a \pi(a|s)\, P(s'|s, a)$
$D_\pi = (I - \beta\, F_\pi^\top)^{-1}\, \rho_0$

`// 3c. Expected features under current policy`
$\bar{\phi}_\pi = \sum_s D_\pi(s) \sum_a \pi(a|s)\, \phi(s, a)$

`// 3d. Feature matching gradient (Eq. 11)`
$g = \bar{\phi}_{\text{data}} - \bar{\phi}_\pi$

`// 3e. Update θ via L-BFGS-B or Adam`
$\theta_{k+1} \leftarrow \theta_k + \alpha\, g$ `// or L-BFGS-B step`

`// 3f. Dual stopping criteria (Gleave & Toyer 2022)`
if $\|g\| < \varepsilon_1$ ***or*** $\|D_{data} - D_\pi\|_\infty < \varepsilon_2$ **then break**

end

`// Step 4: Inference`
$\ell(\hat{\theta}) = \sum_i \log \pi_{\hat{\theta}}(a_i|s_i)$
Bootstrap SEs by resampling individuals and re-estimating

Output: $\hat{\theta}$ (reward weights), V^*, π^*, log-likelihood, bootstrap SEs

Implementation notes. The backward pass (Step 3a) uses the hybrid solver from EconIRL, which starts with successive approximation and switches to Newton-Kantorovich near the fixed point for quadratic convergence. The forward pass (Step 3b) solves the occupancy measure via matrix inversion rather than iterative power method. The dual stopping criterion (Step 3f) follows Gleave et al. (2022), checking both gradient convergence and occupancy distance. L-BFGS-B is the default optimizer because it handles the implicit curvature well, but Adam is available for gradient-based optimization when the Hessian is ill-conditioned.

The deep variant replaces $\theta^\top \phi(s, a)$ with a neural network $r_\psi(s, a)$ following Wulfmeier et al. (2016). The gradient becomes $\nabla_\psi r$ weighted by the occupancy mismatch. This gains flexibility at the cost of interpretability and valid standard errors.

5 Results

Table 1: 5×5 gridworld (25 states, $\beta = 0.95$, 3,000 obs). MCE-IRL and NFXP solve the same MLE.

Metric	MCE-IRL	NFXP	BC
Log-likelihood	−3965.10	−3965.10	−3950.15
Cosine similarity	0.6818	0.1203	—
Policy accuracy (%)	96.0	96.0	76.0
Time (s)	4.0	1.5	0.3

We run MCE-IRL, NFXP, and behavioral cloning on a 5×5 gridworld with 25 states, 5 actions, and three action-dependent features (step penalty, terminal reward, distance weight). Data consists of 3,000 observations at $\beta = 0.95$.

MCE-IRL and NFXP achieve identical policy accuracy (96.0% and 96.0%), confirming that they solve the same MLE. Both substantially outperform behavioral cloning (76.0%), which does not learn from MDP structure. The cosine similarities differ because rewards are identified only up to an additive constant and scale, so the raw parameter vectors may point in different directions while producing the same policy. Policy accuracy is the correct evaluation metric for IRL.

The feature matching gradient (Equation 11) is the key difference from NFXP. NFXP computes the likelihood gradient via implicit differentiation through the Bellman fixed point, which requires solving a linear system at each step. MCE-IRL computes the gradient as a difference between empirical and expected features, which requires the forward occupancy pass but avoids the implicit differentiation. Both approaches have the same computational cost per iteration (one backward pass and one forward solve), but MCE-IRL's gradient has a cleaner interpretation as feature matching.

Code. The companion script `mce_irl_results.py` in this folder generates Table 1. The MCE-IRL estimator is implemented in `src/econirl/estimation/mce_irl.py`. To regenerate:

```
python papers/econirl_package/primers/mce_irl/mce_irl_results.py
```

References

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